

Three key areas / Three operating roles

MULTIMODAL / v1.1

MOBILITY COMMONS / v1.0

Each area has a minimum pilot, accountable owner and stop condition

01

Zhongzhiyuan

enterprise arrival

shuttle merge / loading

pilot / reversible

STOP

No permanent community ban

02

AI Origin

resident daily

human parity / accessible

pilot / reversible

STOP

No home trajectory

03

Dazhongsi

rail + curb

station / cycle / event

pilot / reversible

STOP

No fire/access blockage

服务水平 / SERVICE LEVELS

到达连续 route continuity

换乘可靠 transfer reliability

路缘有序 curb order

申诉闭环 complaint closure

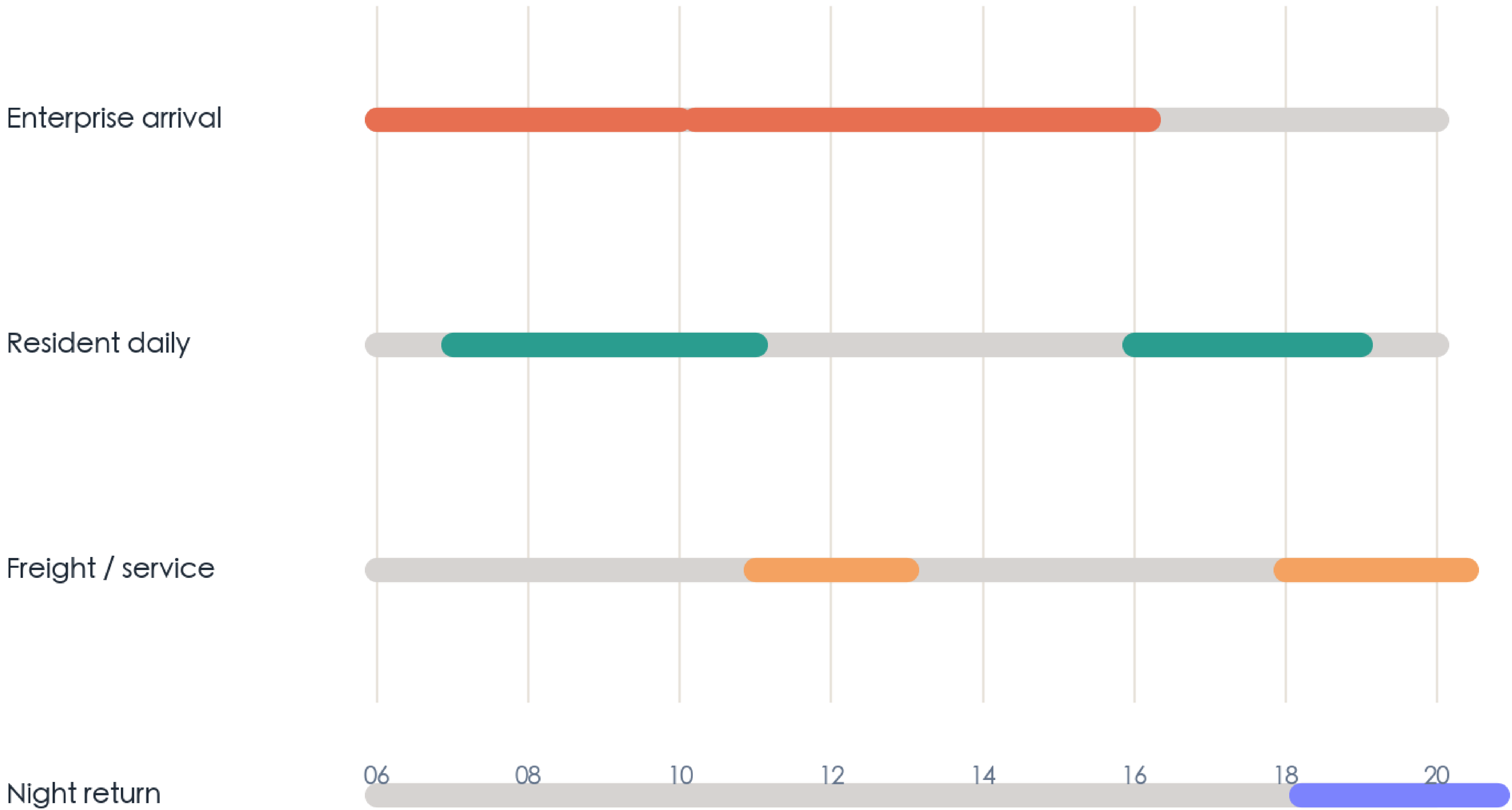
Two-sided demand ledger / Time-windowed land-use

MULTIMODAL / v1.1

MOBILITY COMMONS / v1.0

Enterprise arrival, resident daily access and public curb share one time axis

Time windows are pilot defaults, not a local forecast



数据规则 / data rule

聚合到班次、服务类型和时间窗；不保存个人连续轨迹

Concept land-use base

Enterprise



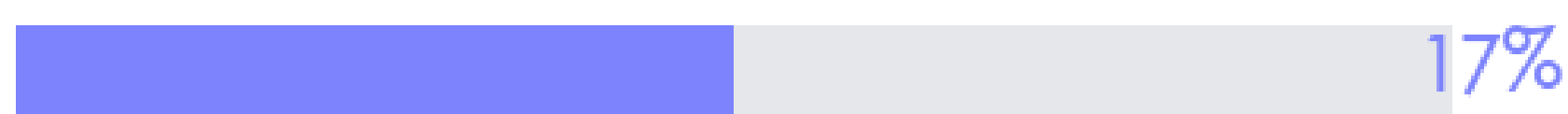
Green



Service



Resident



Key: demand is measured in P0

Regional-scale commute: cover every outbound and return leg

3,122,000 synthetic agents × AM work chain + PM return screen × metro / bus / bicycle / walking-accessible / car / enterprise shuttle · aggregate outputs only

ALL AGENTS PROCESSED

Regional population coverage

3.122M / 100%

Official population reference; not a workforce / OD baseline

O4 capacity-balanced · satisfaction proxy

54.67 → 66.44

+11.77 points · synthetic proxy

O4 capacity-balanced · generalized cost

60.34 → 49.44

-10.90 generalized-cost proxy

MODE MIX / GROUPED DEMAND

Each bar = 3,122,000 AM outbound chains; the PM return screen also conserves the full population; not an observed share

B0 stress case



25.1% 18.1% 8.7% 16.3% 26.3% 5.5%

O4 capacity-bala



30.5% 19.3% 13.5% 20.9% 9.6% 6.2%

Metro

Bus

Bicycle

Walking / accessible

Car

Enterprise shuttle

O4 passes the 1.35× peak-load gate before satisfaction ranking; O3 has a higher paper score but bicycle load reaches 1.57× and is rejected.

Service ledger: train departures / bus departures / bicycle slots / accessible slots / car equivalents / shuttles; passenger-km 34.86 → 28.40M; air remains blocked

SYSTEM READOUT

B0 → O4 · full replay selected · capacity gate ≤ 1.35× · synthetic proxies

P90 travel-time proxy ↓

90 → 60 min

Accessible completion proxy ↑

91.24 → 93.43%

People conflicts / 1000 ↓

5.74 → 3.13

External car inflow ↓

26.95 → 8.47%

Vehicle / service km proxy ↓

9.00 → 3.51M

Peak mode load ↓

1.57× → 1.21×

AGGREGATE ROUTE CHAINS

Every synthetic agent completes one morning chain; only route templates and flow counts survive

Home / boundary

First mile: walk / bike

Backbone: metro / bus

Last mile: accessible

Work / service destination

ONE REGIONAL RUNNER · FIVE READOUT VIEWS

1 regional scale

2 network flow

3 capacity gap

4 resource sensitivity

5 robustness

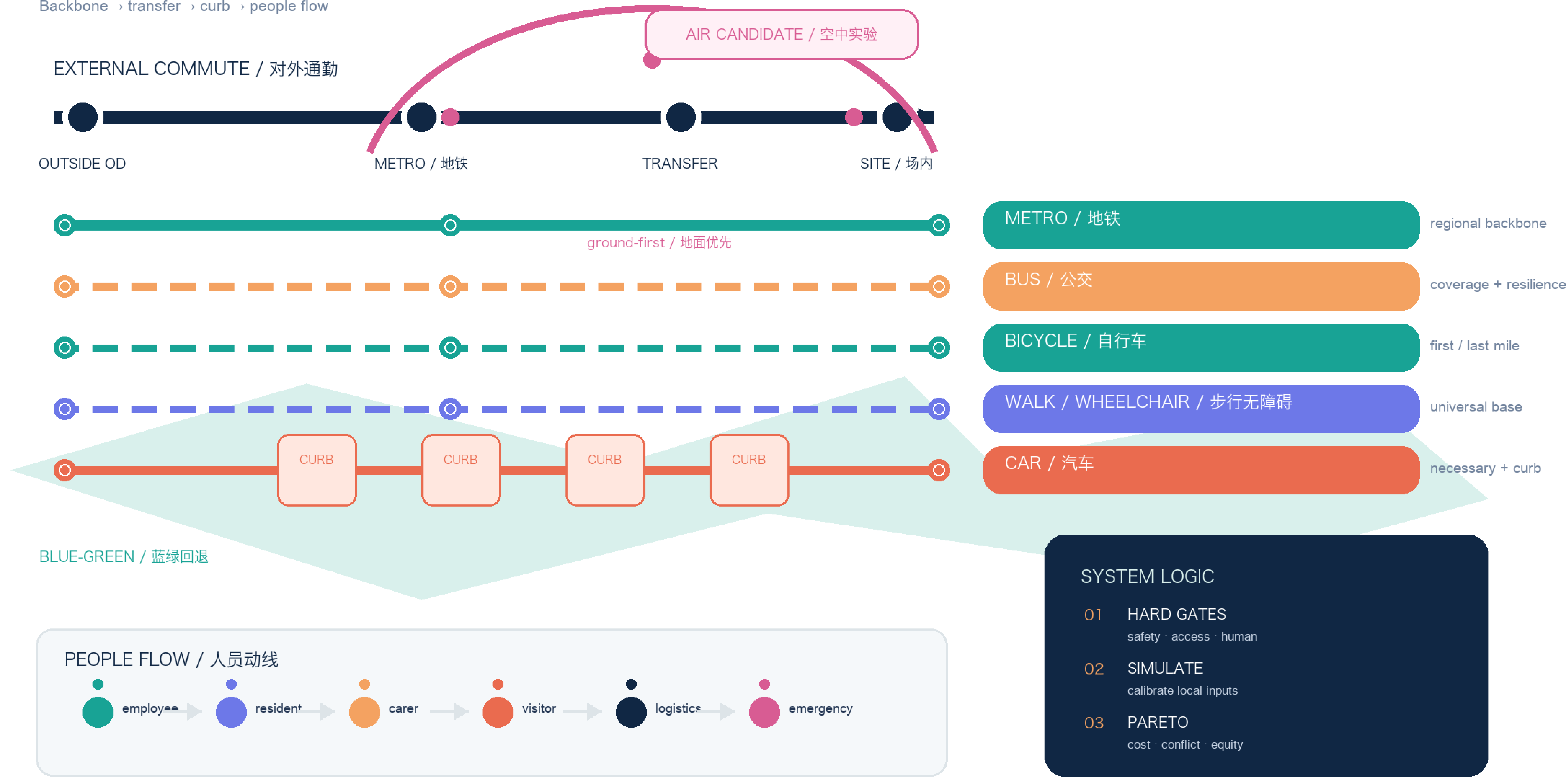
one aggregate evidence chain · not five independent field results

MULTIMODAL MOBILITY LOOP

Five ground modes + one conditional air experiment / people flow stays visible

01 MODE STACK / 方式叠层

Backbone → transfer → curb → people flow



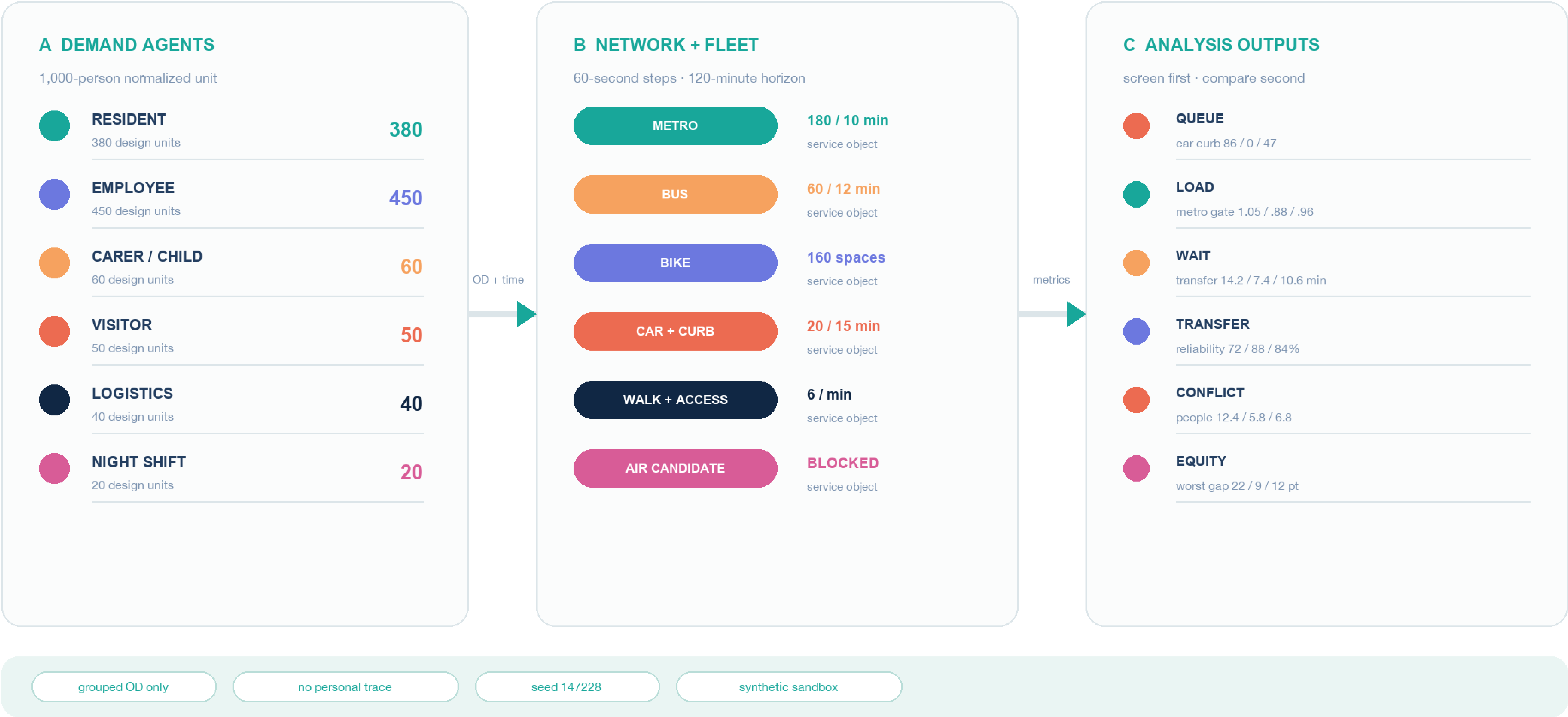
MULTIMODAL MODEL OBJECTS

Residents · vehicles · metro · curb · accessible flow · gated air candidate

JING-ZHANG / MODEL

01 MODEL OBJECTS

Agent-based demand meets capacity, queues and hard gates.



S0 / S1 / S3 values are design sensitivities; replace with dated local calibration before any performance claim.

Source: visual/assets/movement-simulation.json

PROVISIONAL · MODEL SPEC · NOT A BASELINE

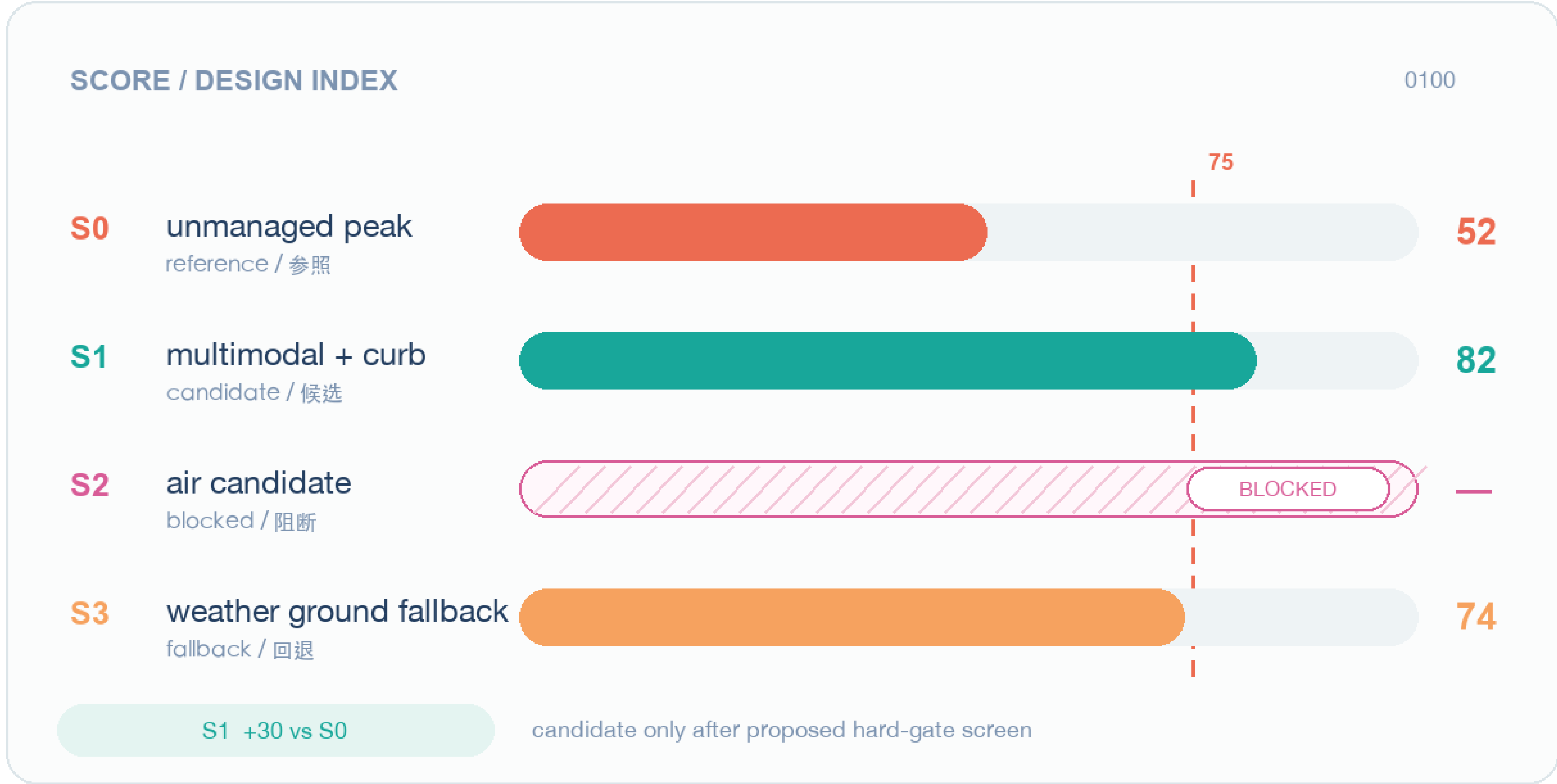
DESIGN-SCENARIO SIMULATION

Transparent sandbox · not a current baseline · gates before scores

P0 ▢ P1 / DESIGN BOARD

01 CANDIDATE EFFICIENCY

0–100 illustrative index · dashed line = 75 suggested threshold



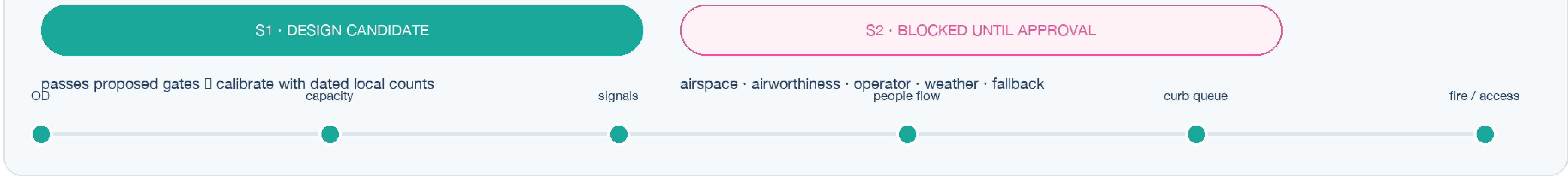
1,000-person design unit

TRADE-OFF READOUT

illustrative values / design inputs

	S0	S1	S2	S3
Generalized cost (lower) index	100	78	—	85
Transfer reliability (higher) %	72%	88%	—	84%
People conflicts (lower) /1000	12.4	5.8	—	6.8
External car inflow (lower) %	42%	25%	—	28%
Worst-group gap (lower) pt	22	9	—	12
Energy index (lower) index	100	77	—	84

02 DECISION & CALIBRATION



Design values only. Replace each row with dated, grouped and locally calibrated evidence before claiming performance.

Source: visual/assets/movement-simulation.json

PROVISIONAL · NOT A BASELINE

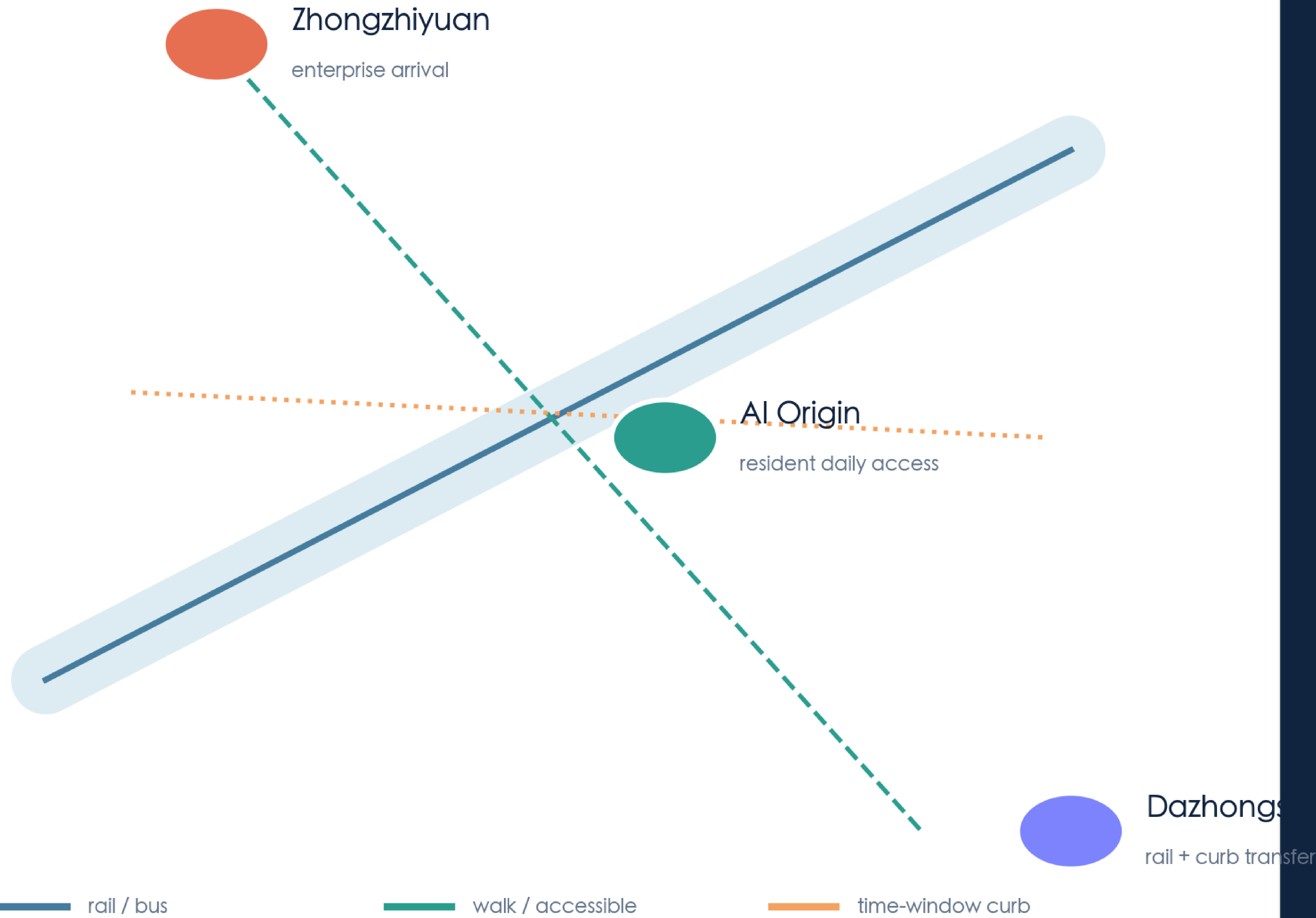
Jing-Zhang Mobility Commons / Enterprise-Resident Mobility

MULTIMODAL / v1.1

MOBILITY COMMONS / v1.0

Three key areas × three connection types × five verification gates

SPATIAL BASE / PROVISIONAL DESIGN AREA



FIVE GATES

- 01 几何 / geometry
official polygon + right-of-way
- 02 需求 / demand
grouped consent + time window
- 03 安全 / safety
accessible + fire continuity
- 04 责任 / operator
SLA + maintenance + complaint
- 05 公平 / equity
human parity + rollback